DSA REFERENCE

# Graph Shortest Path Problem Ladder

A sequenced practice ladder from warm-up to interview stretch.

Expand paths in an order that guarantees a finalized distance cannot later improve.

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STATUS HUMAN REVIEW REQUIRED

DSA / SHORTEST-PATH

DSA

# Graph Shortest Path Problem Ladder

**CORE IDEA**

**Expand paths in an order that guarantees a finalized distance cannot later improve.**

### Recognition cues

+ minimum weighted cost
+ nonnegative edges
+ few negative edges requiring another algorithm
+ path reconstruction

### Implementation sequence

+ classify edge weights
+ choose BFS, 0-1 BFS, Dijkstra, or Bellman-Ford
+ relax edges
+ skip stale heap entries
+ store parents if needed

## Invariant

A popped non-stale Dijkstra entry has the smallest unsettled distance; finalized distances are optimal.

## Worked reasoning

Dijkstra pushes a new pair whenever dist[u] + weight improves dist[v]; stale older pairs are ignored on pop.

## Complexity

Dijkstra with adjacency lists and a binary heap is O((V + E) log V); storage is O(V + E).

## Failure modes

- [ ] using Dijkstra with negative edges
- [ ] marking visited on push
- [ ] overflow in large distances
- [ ] forgetting unreachable nodes

Practice ladder

# Graph Shortest Path Problem Ladder: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

| # | Problem | Deliberate-practice prompt |
|---|---|---|
| 1 | **Network Delay Time** | State the cue, invariant, edge case, and proof before implementation. |
| 2 | **Cheapest Flights Within K Stops** | State the cue, invariant, edge case, and proof before implementation. |
| 3 | **Path With Minimum Effort** | State the cue, invariant, edge case, and proof before implementation. |
| 4 | **Swim in Rising Water** | State the cue, invariant, edge case, and proof before implementation. |
| 5 | **Find the City** | State the cue, invariant, edge case, and proof before implementation. |

## Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

## Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS

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